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import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms

# Define a simple MLP model
class MLP(nn.Module):
    def __init__(self, input_size, hidden_size, num_classes):
        super(MLP, self).__init__()
        self.fc1 = nn.Linear(input_size, hidden_size)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(hidden_size, hidden_size)
        self.fc3 = nn.Linear(hidden_size, num_classes)

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        x = self.relu(x)
        x = self.fc3(x)
        return x

input_size = 32 * 32 * 3 # CIFAR-10 images are 32x32 pixels with 3 channels (RGB)
hidden_size = 256
num_classes = 10
learning_rate = 0.001
num_epochs = 10
batch_size = 100

# Load CIFAR-10 dataset
transform = transforms.Compose([
    transforms.ToTensor(), # Convert to Tensor
    transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5)) # Normalize with mean and std deviation for each channel
])

train_dataset = torchvision.datasets.CIFAR10(root='./data', train=True, download=True, transform=transform)
train_loader = torch.utils.data.DataLoader(train_dataset, batch_size=batch_size, shuffle=True)

Downloading https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz to ./data/cifar-10-python.tar.gz
100%|██████████| 170498071/170498071 [00:04<00:00, 41500012.26it/s]
Extracting ./data/cifar-10-python.tar.gz to ./data

# Initialize the model, loss function, and optimizer
model = MLP(input_size, hidden_size, num_classes) # Initializing the MLP model

# Define the loss function
criterion = nn.CrossEntropyLoss() # Using Cross Entropy Loss for classification tasks

# Define the optimizer
optimizer = optim.Adam(model.parameters(), lr=learning_rate) # Adam optimizer with the given learning rate

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# Training the model
for epoch in range(num_epochs):
    total_correct = 0
    total_samples = 0
    for i, (images, labels) in enumerate(train_loader):
        images = images.view(-1, 32 * 32 * 3) # Flatten the images from 32x32x3 to 1x(32*32*3)

        # Forward pass
        outputs = model(images)
        loss = criterion(outputs, labels)

        # Backward and optimize
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        # Accuracy calculation
        _, predicted = torch.max(outputs, 1)
        total_samples += labels.size(0)
        total_correct += (predicted == labels).sum().item()

    # Print training statistics
    if (i+1) % 100 == 0:
        print(f'Epoch [{epoch+1}/{num_epochs}], Step [{i+1}/{len(train_loader)}], Loss: {loss.item():.4f}, Accuracy: {(total_correct/tot

print("Training finished")

# Save the trained model
torch.save(model.state_dict(), 'mlp_cifar10_model.pth')

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Epoch [1/10], Step [100/500], Loss: 1.7702, Accuracy: 35.17%
Epoch [1/10], Step [200/500], Loss: 1.5222, Accuracy: 37.92%
Epoch [1/10], Step [300/500], Loss: 1.8210, Accuracy: 39.66%
Epoch [1/10], Step [400/500], Loss: 1.5456, Accuracy: 41.05%
Epoch [1/10], Step [500/500], Loss: 1.4434, Accuracy: 42.14%
Epoch [2/10], Step [100/500], Loss: 1.4900, Accuracy: 48.52%
Epoch [2/10], Step [200/500], Loss: 1.3975, Accuracy: 49.39%
Epoch [2/10], Step [300/500], Loss: 1.3641, Accuracy: 49.76%
Epoch [2/10], Step [400/500], Loss: 1.4561, Accuracy: 49.94%
Epoch [2/10], Step [500/500], Loss: 1.5047, Accuracy: 50.04%
Epoch [3/10], Step [100/500], Loss: 1.3539, Accuracy: 54.54%
Epoch [3/10], Step [200/500], Loss: 1.2496, Accuracy: 53.99%
Epoch [3/10], Step [300/500], Loss: 1.3534, Accuracy: 54.15%
Epoch [3/10], Step [400/500], Loss: 1.2801, Accuracy: 54.15%
Epoch [3/10], Step [500/500], Loss: 1.3755, Accuracy: 54.09%
Epoch [4/10], Step [100/500], Loss: 1.2889, Accuracy: 58.19%
Epoch [4/10], Step [200/500], Loss: 1.0958, Accuracy: 58.17%
Epoch [4/10], Step [300/500], Loss: 1.4072, Accuracy: 57.91%
Epoch [4/10], Step [400/500], Loss: 1.4145, Accuracy: 57.60%
Epoch [4/10], Step [500/500], Loss: 1.1884, Accuracy: 57.47%
Epoch [5/10], Step [100/500], Loss: 0.9837, Accuracy: 61.69%
Epoch [5/10], Step [200/500], Loss: 1.1671, Accuracy: 60.89%
Epoch [5/10], Step [300/500], Loss: 1.1769, Accuracy: 60.52%
Epoch [5/10], Step [400/500], Loss: 1.1759, Accuracy: 60.40%
Epoch [5/10], Step [500/500], Loss: 0.9864, Accuracy: 60.26%
Epoch [6/10], Step [100/500], Loss: 1.1723, Accuracy: 63.89%
Epoch [6/10], Step [200/500], Loss: 1.0992, Accuracy: 63.59%
Epoch [6/10], Step [300/500], Loss: 0.8244, Accuracy: 63.15%
Epoch [6/10], Step [400/500], Loss: 0.8086, Accuracy: 63.03%
Epoch [6/10], Step [500/500], Loss: 1.0029, Accuracy: 62.89%
Epoch [7/10], Step [100/500], Loss: 1.0915, Accuracy: 67.13%
Epoch [7/10], Step [200/500], Loss: 0.8541, Accuracy: 66.53%
Epoch [7/10], Step [300/500], Loss: 1.2288, Accuracy: 65.82%
Epoch [7/10], Step [400/500], Loss: 0.8743, Accuracy: 65.45%
Epoch [7/10], Step [500/500], Loss: 0.9443, Accuracy: 65.24%
Epoch [8/10], Step [100/500], Loss: 0.8092, Accuracy: 69.37%
Epoch [8/10], Step [200/500], Loss: 0.8235, Accuracy: 68.16%
Epoch [8/10], Step [300/500], Loss: 0.8685, Accuracy: 67.98%
Epoch [8/10], Step [400/500], Loss: 0.8329, Accuracy: 67.71%
Epoch [8/10], Step [500/500], Loss: 0.9120, Accuracy: 67.41%
Epoch [9/10], Step [100/500], Loss: 0.8062, Accuracy: 72.13%
Epoch [9/10], Step [200/500], Loss: 0.5511, Accuracy: 70.85%
Epoch [9/10], Step [300/500], Loss: 0.8347, Accuracy: 70.63%
Epoch [9/10], Step [400/500], Loss: 0.9584, Accuracy: 70.22%
Epoch [9/10], Step [500/500], Loss: 1.0068, Accuracy: 69.80%
Epoch [10/10], Step [100/500], Loss: 0.6377, Accuracy: 73.79%
Epoch [10/10], Step [200/500], Loss: 0.7286, Accuracy: 73.00%
Epoch [10/10], Step [300/500], Loss: 0.8148, Accuracy: 72.79%
Epoch [10/10], Step [400/500], Loss: 0.7133, Accuracy: 72.42%

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Epoch [10/10], Step [500/500], Loss: 0.7955, Accuracy: 72.04%
Training finished

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# Define the model architecture and load the saved model
model = MLP(input_size, hidden_size, num_classes) # Define the model with the same architecture
model.load_state_dict(torch.load('mlp_cifar10_model.pth')) # Load the saved model state

# Set the model to evaluation mode
model.eval()

# Load CIFAR-10 test dataset
test_dataset = torchvision.datasets.CIFAR10(root='./data', train=False, download=True, transform=transform)
test_loader = torch.utils.data.DataLoader(test_dataset, batch_size=batch_size, shuffle=False)

# Test the model on the test data
correct = 0
total = 0
with torch.no_grad():
    for images, labels in test_loader:
        images = images.view(-1, 32 * 32 * 3) # Flatten the images from 32x32x3 to 1x(32*32*3)
        outputs = model(images)
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

# Print the accuracy results
print(f'Accuracy on the test set: {(100 * correct / total):.2f}%')
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Files already downloaded and verified
Accuracy on the test set: 53.19%